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A Generalizable Topological Framework for Adaptive Anastomotic Patterning via Residual Prime Manifolds

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Abstract

This study introduces a novel non-rigid registration paradigm governing the surgicobotanical application of 'surfboard' excisions in complex anastomosis. By modeling stochastic deformation fields as Wishart distributions interacting with biological elliptic properties characterized by generalized continued fractions, we constrain the Coherent Point Drift (CPD). The error surfaces are parameterized using residual differential equations grounded in finite properties of prime numbers to guarantee boundary continuity during skin or vascular grafting.

1. Coherent Point Drift (CPD) & Wishart Noise

We model the moving lattice Y transitioning towards the target dermal lattice X using a Gaussian Mixture Model (GMM) augmented by Wishart distributed noise simulating biological elasticity.

$$P(\mathbf{x}_i|\mathbf{Y}) = \sum_{j=1}^{M} \frac{1}{M \cdot (2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|\mathbf{x}_i - \mathcal{T}(\mathbf{y}_j, \boldsymbol{\theta})\|^2}{2\sigma^2}\right)$$

$$W(\Sigma|S, \nu) = \frac{|\Sigma|^{(\nu - D - 1)/2}}{2^{\nu D/2}|S|^{\nu/2}\Gamma_D(\frac{\nu}{2})} \exp\left(-\frac{1}{2}\mathrm{Tr}(S^{-1}\Sigma)\right)$$

Where S denotes the target spatial covariance structure dictating the allowable deformation limits of the viscoelastic tissue in the surfboard anastomosis.

2. Elliptic Continuation & Prime Residual Diff. Eq.

To resolve pathological singularities at the incision vertices, we apply a continued fraction representation to the elliptic integrals governing the geometry:

$$\mathcal{E}(\phi, k) = \frac{a_1}{b_1 + \frac{a_2}{b_2 + \frac{a_3}{b_3 + \dots}}}$$

The residual tracking of the displacement field $u(t)$ across modular prime properties is defined as:

$$\frac{\partial u}{\partial t}(x) \equiv \sum_{p \in \mathbb{P}} \left(\nabla^2 u + p \cdot \mathcal{F}(x)\right) \pmod{p}$$

Conclusion

These models define a generalized paradigm for simulated physiological anastomosis, verifying spatial coherence through highly stochastic topological limitations.